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Multi-key spatio-temporal chaotic image encryption and retrieval scheme

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Abstract To protect privacy, many users opt to encrypt images prior to outsourcing them to cloud service platforms (CSPs). However, this encryption process results in the loss of image features and subsequent inability to retrieve them. We propose an image encryption and retrieval algorithm, which ensures that privacy is not leaked in both the upload and retrieval stages. First, overcoming the insufficient security and degradation in chaotic systems, we introduce the time-varying functions, and unscented Kalman filter to improve the non-adjacent coupled map lattice complexity and security. Secondly, considering the encryption efficiency, we compress the plaintext image to reduce the time of the encryption phase and improve the overall encryption speed. Finally, we use the locally sensitive hash (LSH) for feature vector dimensionality reduction to improve the retrieval efficiency and perform a secondary LSH on the reduced feature vector to form a new hash-key retrieval structure in the generate index phase, which improves the retrieval efficiency. The experimental results prove that our proposed encryption algorithm can meet the image encryption algorithm's high retrieval accuracy and multi-user without revealing privacy.

Keywords Encrypted image retrieval · Image encryption · Chaos degradation · Multiple-time-vary NA-CML

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1 Introduction

With the proliferation of smart devices such as mobile phones, laptops, and surveillance cameras, there has been a significant increase in the volume of multimedia data. Consequently, data owners are increasingly resorting to cloud outsourcing [1]. However, due to concerns regarding potential breaches in data privacy, data owners have started encrypting their data before uploading it to the cloud for enhanced security during both upload and storage processes. Nevertheless, this encryption process results in the loss of retrieval characteristics for the encrypted data; thus rendering many plaintext-based image retrieval schemes unsuitable for direct application within the ciphertext space. As a result, data owners are required to download all encrypted data and decrypt it in order to retrieve

specific images they desire [2]. This situation poses considerable inconvenience for both cloud servers and data owners [3–5].

Many scholars have put forward a variety of targeted solutions from the perspectives of protecting image privacy and retrieving encrypted images.

In the image encryption, the researchers noted that unlike text data, image data has strong pixel correlation and data redundancy. Therefore, existing symmetric encryption schemes may not be effective in eliminating these problems [6–12]. Majid et al. proposed a chaos-based encryption algorithm that can encrypt text and image files, compute the encryption key according to the DNA rules, and finally achieve the one-time pad (OTP) [6]. Wu et al. used cascading multiple chaotic maps as the driving system of Coupled Lattice Map (CML) to expand the CML parameter range and increase the encryption range. However, the complexity is too high in the implementation process, and it is not suitable for large-scale encryption [7]. Wang et al. proposed a new chaotic system that provides pseudo-random sequences for the encryption algorithm [8]. Zhang et al. proposed a new spatio-temporal chaotic system combining linear and nonlinear systems, avoiding the CML periodic window problem and using DNA sequences for diffusion operations in the image encryption process [9].

In the aspect of encrypted image retrieval, researchers use image feature vector as index, and use symmetric or asymmetric encryption scheme to encrypt it, and achieve retrieval through trap function. Many searchable encryption schemes have been proposed for textual and medical data [13, 14], but the existing schemes are not well adapted to encrypted image retrieval. There has been much discussion on retrieving encrypted images quickly and accurately without compromising image privacy. However, most of the existing image retrieval schemes are single-key ciphertext retrieval mechanisms, which are not well adapted to real application scenarios. The sharing in a single-key mechanism can easily cause the leakage of user keys, leading to the leakage of user privacy. In addition, most schemes also have disadvantages, such as low retrieval accuracy and efficiency.

Both chaotic image encryption and encrypted image retrieval encounter challenges due to the limited computing power and storage capacity of smart devices. Therefore, it is crucial to consider the actual capabilities of smart devices while ensuring security. However, despite its strong performance in engineering indicators, the chaotic encryption algorithm often suffers from high complexity and degradation issues during the specific encryption process, resulting in a narrow encryption range and compromised security. Additionally, when designing an encrypted image retrieval scheme, assuming that the data owner and retriever are the same person may lead to key leakage problems and potential disclosure of private data when sharing images.

2 Motivation

2.1 Image encryption based on chaos

As the most effective way to protect image privacy, encrypted images have been proposed to design image encryption algorithms based on chaos theory [15–17]. Compared with the AES, DES, and 3DES encryption schemes, chaos-based image encryption uses a permutation-diffusion model to encrypt images, which can overcome the problem that traditional algorithms (e.g., AES, DES) cannot eliminate pixel correlation when encrypting images [8, 18]. However, the security of chaotic image encryption has been a concern in recent years [19, 20], as chaotic systems suffer from chaotic degradation with the increase in the number of iterations and the influence of the computational accuracy of digital circuits [21], leading to a reduction in the range of parameters that can be easily broken by brute force [13–15]. In order to address the issue of degradation in chaos, a multitude of solutions have been proposed by researchers. Such as Lv [22] introduced time lag functions into the coupled mapping lattice by introducing the time dimension into the one-dimensional chaotic mapping, based on which the newly proposed chaotic mapping can perturb the system in the time dimension, which increases the complexity of the system. Wang [15] designed a full chaotic spatio-temporal chaotic system and applied it to image encryption. Wang [17] proposed encrypting the image using the truth table and improved the corresponding none-dimensional chaotic system to compensate for some periodic window problems.

None of the above chaos-based image encryption schemes have proposed an effective solution to the chaotic degradation problem [23]. They just simply couple one-dimensional chaotic systems, and although

the output in no blank window, the complexity of the newly proposed systems is not significantly improved, and the security is all dependent on the complexity of the original one-dimensional system, which many researchers have proved that these encryption schemes are deficient in security [20, 24]. Therefore, we propose a Multi-time-varying Spatio-temporal chaotic system to effectively address the issue of chaotic degradation.

2.2 Image-based searchable encryption

The statistical properties of encrypted images, such as pixel-adjacency correlation and histogram derived from the original image, are effectively eliminated during the encryption process [25]. This way can ensure that the user's privacy is not leaked, making it possible to store the encrypted image directly to the remote cloud server. However, how can users efficiently and securely retrieve the enormous number of encrypted images with the increasing number of encrypted images?

Many scholars have proposed various ciphertext-based retrieval schemes to retrieve encrypted images [13, 14, 26–29]. In the order-preserving encryption (OPE) and comparable encryption (CE) schemes [30–32], smart devices are required to extract picture features, create image indexes, and encrypt original images. However, smart devices have limited processing power and must upload significant amounts of data, making it challenging to meet user needs.

Furthermore, in order to extract image feature vectors, OPE and CE methods typically use scale-invariant feature transform (SIFT) [33, 34], speed up robust features (SURF) [35], and oriented fast and rotated brief (ORB) [36] which may have algorithm robustness issues and result in low ciphertext retrieval accuracy. Hsu et al. [37] proposed the privacy-protect SIFT (PP-SIFT) scheme to extract image features in the ciphertext domain, and the extraction accuracy of PP-SIFT and SIFT is approximately the same, while in the ciphertext image retrieval experiments, the retrieval accuracy of PP-SIFT is only slightly lower than that of SIFT. Meanwhile, during the code implementation process of PP-SIFT, after modifying the calculation rules of SIFT, it fails to accurately compute the desired result, resulting in divergent calculation outcomes for PP-SIFT. Zou proposed a cryptographic search scheme based on comparable encryption and order-preserving encryption for content-based image retrieval [32]. This scheme avoids homomorphic encryption, which greatly reduces the client's computation and improves the accuracy of the fuzzy search. The combination of homomorphic encryption and asymmetric scalar-product-preserving encryption (ASPE) was proposed by Li et al. to encrypt image features [38]. Zhang et al. [39] used an additive secret-sharing technique to encrypt images and introduced joint learning to propose an encrypted image comparison protocol. Nonetheless, this scheme requires multiple rounds of communication between the two parties, which cannot guarantee that the user to be online for a long time in real life.

In order to realize the high efficiency of image encryption and retrieval, we propose a multi-key scheme of image encryption and retrieval based on spatio-temporal chaotic system. The image encryption and retrieval architecture are shown in Fig. 1. Cloud1 stores the encrypted images, while Cloud2 is responsible for storing the encrypted image indexes. After the feature vectors are extracted from the image, the image owner uses it to encrypt images. The image owner needs to upload both the encrypted images and the encrypted keys. In order to protect image privacy from leakage, Cloud1 and Cloud2 cannot be any communication between these two cloud servers. Therefore, we introduce the key management center (KMC) server responsible for authenticating the query user identity and key distribution. Under the semi-honest model of the cloud platform, to protect image privacy from being leaked, we store the image retrieval index and the image in the different cloud platforms, and these two cloud platforms cannot communicate in any form. Even if either cloud platforms are hacked or spied on, the privacy of the images is assured to remain intact. In order to solve the problem of low retrieval rate, we propose to use SIFT to extract local image features, and the local features of the image are extracted and used as the index of the encrypted image to improve the retrieval accuracy of the encrypted image. We use the Local Sensitive Hash to compress the 128-dimensional feature vector while

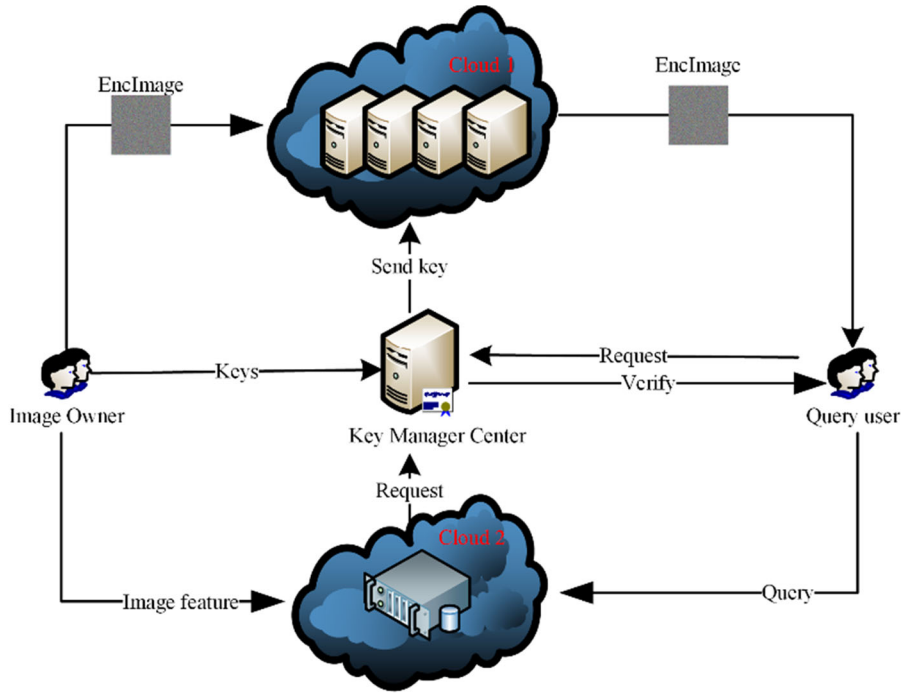


Fig. 1 System architecture

performing a second hash operation to improve the retrieval operation in a massive image database for low retrieval efficiency.

The main contributions of this manuscript are summarized as follows:

To overcome the period window problem arising from the iteration of the Logistic system in digital circuits, we use the UKF to compensate for the error of the Logistic system by using the new Logistic as a local mapping of the non-adjacent coupled mapping lattice. Also, introduce the time lag function to increase the complexity of the non-adjacent coupled mapping lattice in the time dimension.

In the image encryption phase, we use the perceptual compression algorithm to reduce the computation in the encryption phase and improve the overall encryption speed. Meanwhile, we use the image feature vector as the encryption key to achieve a one-time pad (OTP) encryption effect.

The organization of the rest of this paper is as follows:

Section 3 describes the implementation of Multi-time-vary NA-CML; Sect. 4 introduces the encryption algorithm and the ciphertext index construction process. Section 5 is the performance and security analysis; Sect. VI is the conclusion.

3 Multi-time-vary NA-CML

3.1 Multi-time-vary non-adjacent coupled map lattice

The coupling pattern of the non-adjacent coupled mapping lattice undergoes random variations, extending from i-1, i, and $i + 1$ to all the lattices of NA-CML and increasing the complexity of NA-CML. Moreover, the NA-CML retains the characteristics of the CML, such as discrete-time, discrete-space, and continuous state variables [40].

NA-CML is defined as follows:

$$x_{n+1}(i) = (1-\varepsilon)f(x_n(i)) + \frac{\varepsilon}{2}(f(x_n(j)) + f(x_n(k))) \tag{1}$$

$$\begin{bmatrix} j \\ k \end{bmatrix} = \begin{bmatrix} 1 & p \\ q & pq+1 \end{bmatrix} \begin{bmatrix} i \\ i \end{bmatrix} \bmod L \tag{2}$$

where $1 \le i,j,k \le L$ $1 \le i,j,k \le L$ $1 \le i,j,k \le L, L$ is the maximum number of lattices, n is the time index, ε is the coupling coefficient, and p, q are positive integers. The f is a chaotic function. Generally, the f is the Logistic map. The edge conditions satisfy $x_n(L) = x_n(0)$.

It is significant to note that NA-CML modifies the CML couple method and enhances the spatial dimension complexity of the system. In this paper, we introduced the time-vary delay function [22, 41] to improve the time dimension complexity of the NA-CML. The time-vary delay function is when some past moments can affect the present state. Time-varying systems cause complex dynamical behavior as time increases, and this behavior creates uncertainty, diversity, and superposition in the system.

Multi-time-vary NA-CML is as follows:

$$x_{n+1}(i) = (1-\varepsilon)f(x_n(i)) + \frac{\varepsilon}{2}\left[f\left(x_{n-\tau h(n)}(j)\right) + f\left(x_{n-\tau g(n)}(k)\right)\right] \tag{3}$$

$$\tau_1(n+1) : \tau h(n+1) = \left\lfloor n \left| \cos(w \arccos(2\frac{n}{N} - 1) \right| \right\rfloor \tag{4}$$

$$\tau_2(n+1) : \tau g(n+1) = \left\lfloor \left| \frac{n^2}{N} \sin\left(\pi \frac{n}{N}\right) \right| \right\rfloor \tag{5}$$

where $1 \le i,j,k \le L$ $1 \le i,j,k \le L$ $1 \le i,j,k \le L, L$ is the maximum number of lattices, n is the time index, ε is the coupling coefficient, and p, q are positive integers. The f is a chaotic function. Generally, the f is the Logistic map. The edge conditions satisfy $x_n(L) = x_n(0)$. The parameters i,j,k are generated by Eq. (2). The N is the total time delay, and n is the current time and $w \ge 2$.

3.2 Full chaos multi-time-vary NA-CML

We use Logistic map and piecewise linear chaotic map (PWLCM) to address the period window [42], restricted parameter range [43], and unequal distribution of sequences [17, 42].

The Logistic mapping is defined as follows:

$$f(x) : x_{i+1} = \mu x_i(1 - x_i), \mu \in (0, 4] \tag{6}$$

At $\mu \in [3.9, 4]$, the Logistic map is in a chaotic state.

The mathematical definition of PWLCM is as follows:

$$F(x,Q) = \begin{cases} \dfrac{x}{Q}, & 0 \le x < Q \\ \dfrac{(x-Q)}{(0.5-Q)}, & Q \le x < 0.5 \\ \dfrac{(1-Q-x)}{(0.5-Q)}, & 0.5 \le x < 1-Q \\ \dfrac{(1-x)}{Q}, & 1-Q \le x \le 1 \end{cases} \tag{7}$$

Q is the control parameter of PWLCM, and $x \in [0, 1]$.

The mathematical definition of LP is in Eq. (8):

$$x_{i+1} = \begin{cases} \bmod(\mu x_i(1-x_i) + 8x_i/\mu, 1), 0 \le x_i < 0.125\mu \\ \bmod(\left(\dfrac{\mu x_i(1-x_i) + x_i - 0.125\mu}{0.5 - 0.125\mu}, 1\right), 0.125\mu \le x_i < 0.5 \\ \bmod(\mu x_i(1-x_i) + 8(1-x_i)/\mu, 1), 0.5 \le x_i < 1 - 0.125\mu \\ \bmod(\left(\dfrac{\mu x_i(1-x_i) + 1 - x_i - 0.125\mu}{0.5 - 0.125\mu}, 1\right), 1 - 0.125\mu \le x_i \le 1 \end{cases} \tag{8}$$

where $\mu \in (0, 4]$, the LP is in a chaotic state, and $x \in [0, 1]$.

Due to the computational accuracy of computers or digital circuits [44–46], digital chaos systems usually undergo dynamical degradation, resulting in digital chaos systems failing to enter the chaotic state. To overcome the dynamical degradation, we used unscented Kalman filter to compensate for the quantization error resulting from the local chaos map in spatio-temporal systems and by adding time lag perturbation in the iterative process of digital chaos circuits to make the chaotic system more resistant to attacks such as phase space reconstruction [44]. The unscented Kalman filter combines the unscented transform (UT) and the standard Kalman algorithm, which makes the nonlinear system applicable to the traditional Kalman system under the assumption of linearity of the unscented transform, mainly used for system state prediction and update. Due to the advantages of the UKF in prediction, such as the predicted data being close to the ideal data in a mathematical sense, the improved system can have almost all properties similar to the original chaotic system.

3.3 Multiple-time-vary NA-CML

Our pseudorandom number generator is described as follows:

We set the lattice value to $LATTICE = 10$ and the number of iterations $Times = 1000$. The lattices initial value is x = [0.1622, 0.7943, 0.3112, 0.5285, 0.1656, 0.6020, 0.2630, 0.6541, 0.6892, 0.7482]. Our algorithm flowchart is shown in Fig. 2. The bifurcation and the distribution of lattice states under different conditions of multi- time-vary NA-CML are shown in Figs. 3 and 4, respectively.

Correlation analysis measures the degree of correlation between two pseudo-random sequences. We measure the closeness between two sequences using the auto-correlation function, which describes the degree of dependence between two different state values of any pseudo-random sequence. Ideally, the auto-correlation function should be an incremental function, similar to the δ function. We show auto-correlation result of Multi-time-vary NA-CML in Fig. 5.

The waveform of the auto-correlation in Fig. 5 is very similar to the δ function, proving that the degree of dependence between the two sequences is very low. All these show that the Multi-time-vary NA-CML is very close to the ideal value, which proves Multi-time-vary NA-CML can be applied in the cryptography.

Step 1 The observation map and state map of the LP map are defined as in Eq. and

$$x_{i+1} = B(LP(u, x_i), P) + w_k \tag{9}$$

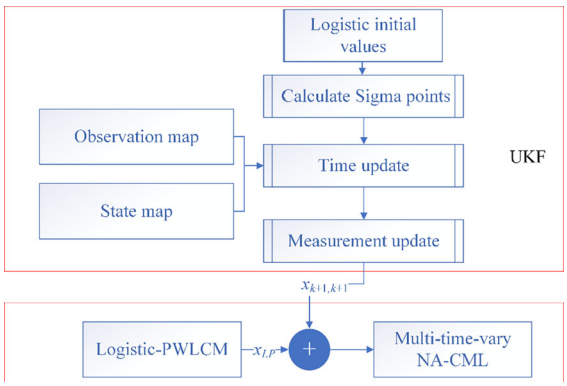


Fig. 2 The flowchart of Multi-time-vary NA-CML

$$y_i = B(x_i, P) + v_k \tag{10}$$

$$B = \frac{\lfloor x \times 2^P \rfloor}{2^P} \tag{11}$$

where B is a quantization function [45], this quantization function corresponds to three rounding operations, ceil, floor, and round, leading to the degeneracy of the discrete phase space. The w_k and v_k are zero-mean, and the corresponding covariances are denoted by Q_k and R_k, respectively. The P is the calculation accuracy, which $P = 8$ in the state map and $P = 16$ in the observation map.

Step 2 Let the LP iterate $n = 1000$ times, the value after the iteration is x_0, and calculate the mean $\hat{x_0}$ and variance P_{00} of x_0.

$$\begin{cases} \hat{x_0} = E(x_0) \\ P_{00} = E(x_0 - \hat{x_0})(x_0 - \hat{x_0})^T \end{cases} \tag{12}$$

Step 3 Selecting Sigma sampling points. Suppose there are L sampling points and $L = 2n$, then there are $2n + 1$ Sigma sampling points. The sampling points are calculated using Eq. (13):

$$\begin{cases} \chi_0 = \hat{x_0} \\ \chi_i = \hat{x_0} + (\sqrt{P_{x,x}(n+k)})_i, i = 1, 2, ..., n \\ \chi_{i+n} = \hat{x} - (\sqrt{P_{x,x}(n+k)})_i \end{cases} \tag{13}$$

The corresponding mean weights and variance weights are:

$$W_i^m = W_i^c = \begin{cases} k/(n+k), i = 0 \\ 1/2(n+k), i \neq 0 \end{cases} \tag{14}$$

where k is the scale factor of the distance between the mean $\hat{x_0}$ and the Sigma sampling points. $\sqrt{(n+k)P_{x,x}}$ is the ith row or column of the matrix's square root matrix $(n+k)P_{x,x}$ after Cholesky decomposition.

Using sampling points for nonlinear transfer of Logistic map:

$$\gamma_{k+1|k}^i = f(\chi_i) \tag{15}$$

The $f()$ is the state map of the Logistic map.

Calculate the mean and variance of the state predictions:

$$\begin{cases} \hat{x}_{k+1,k} = \sum_{i=0}^{2n} (W_i^m)\gamma_{k+1,k}^i \\ P_{k+1,k} = \sum_{i=0}^{2n} (W_i^c)(\gamma_{k+1,k}^i - \hat{x}_{k+1,k})(\gamma_{k+1,k}^i - \hat{x}_{k+1,k})^T + Q_k \end{cases} \tag{16}$$

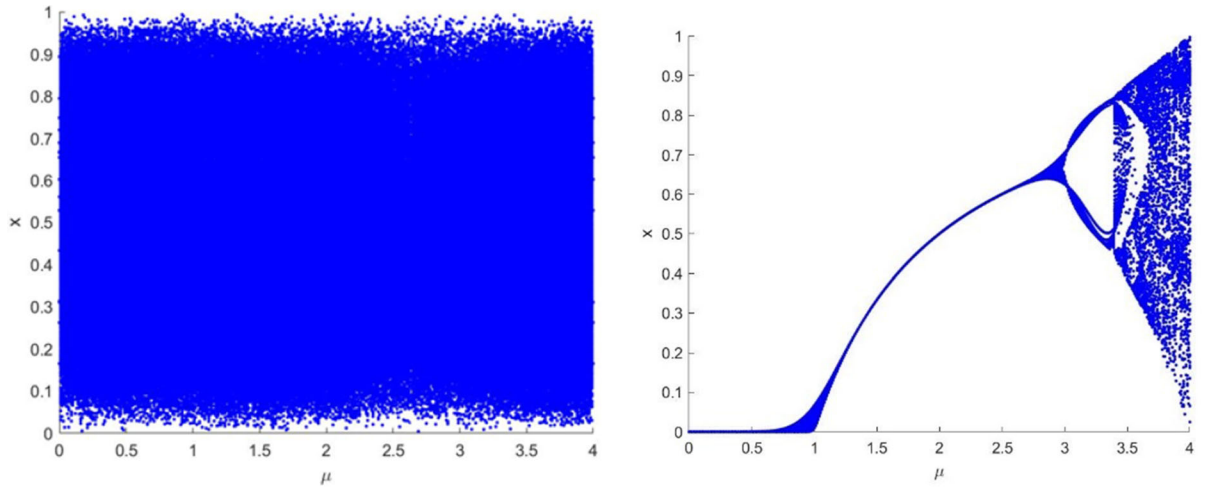


Fig. 3 Bifurcation

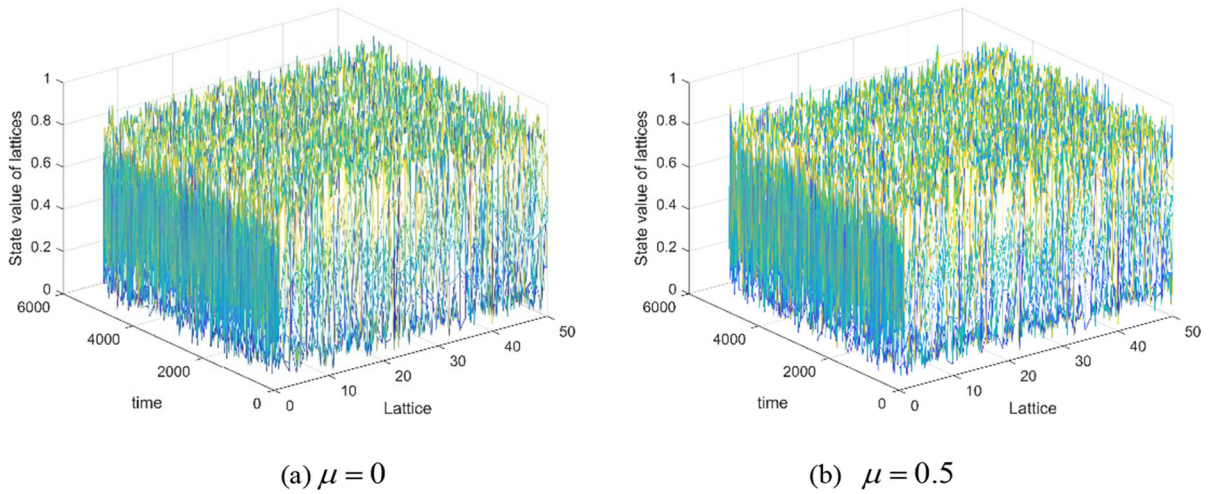


Fig. 4 The state of multi-time-vary NA-CML with different conditions

Step 4 Measurement Updates. The $2n + 1$ sampling points ς_i, and the corresponding weights $(W_i^m)'$ and $(W_i^c)'$ can be obtained according to $\hat{x}_{k+1,k}, P_{k+1,k}$, and the sampling strategy formula.

Using sampling points for nonlinear transfer:

$$\zeta_{k+1,k}^i = h(\varsigma_i) \tag{17}$$

The $h()$ is the observation map of the LP map.

Use Eq. (18) to calculate the mean, variance, and covariance:

$$\begin{cases} \hat{z}_{k+1,k} = \sum_{i=0}^{2n} (W_i^m)' \zeta_{k+1,k}^i \\ P_{zz,(k+1,k)} = \sum_{i=0}^{2n} (W_i^c)' (\zeta_{k+1,k}^i - \hat{z}_{k+1,k})(\zeta_{k+1,k}^i - \hat{x}_{k+1,k})^T + R_{k+1} \\ P_{xz,(k+1,k)} = \sum_{i=0}^{2n} (W_i^c)' (\gamma_{k+1,k}^i - \hat{x}_{k+1,k})(\zeta_{k+1,k}^i - \hat{z}_{k+1,k})^T \end{cases} \tag{18}$$

The filter gain K_{k+1}, the state estimate, and the estimated variance can be calculated from z_{k+1} at the moment $k + 1$.

$$\begin{cases} \hat{x}_{k+1,k+1} = \hat{x}_{k+1,k} + K_{k+1}(z_{k+1} - \hat{z}_{k+1,k}) \\ K_{k+1} = P_{xz,(k+1,k)}(P_{zz,(k+1,k)})^{-1} \\ P_{(k+1,k+1)} = P_{k+1,k} - K_{k+1} P_{zz,(k+1,k)} K_{k+1}^T \end{cases} \tag{19}$$

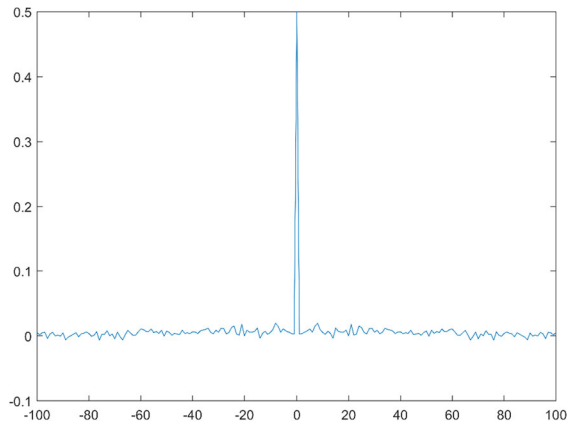


Fig. 5 Auto-correlation

Step 5 In addition, the predicted chaotic system is perturbed under UKF conditions considering the effect of limited accuracy and further enhancement of applicability in cryptography. Eventually, we can obtain the full chaos multi-time-vary NA-CML.

$$\begin{cases} a = \mod(x_n(i) + \hat{x}_{k+1,k+1} + x_{LP}, 1) \\ b = \mod(x_{n-\tau h(n)}(j) + \hat{x}_{k+1,k+1} + x_{LP}, 1) \\ c = \mod(x_{n-\tau g(n)}(k) + \hat{x}_{k+1,k+1} + x_{LP}, 1) \\ x_{n+1}(i) = (1-\varepsilon)f(a) + \dfrac{\varepsilon}{2}[f(b) + f(c)] \end{cases} \tag{20}$$

4 The proposed image encryption and retrieval algorithm

4.1 Encryption processing

The size of the plain image P is $M \times N$. M and N are the image's length and breadth, respectively.

Step 1 The feature points of the plain image P are extracted using the SIFT, and the feature matrix is denoted as F.

$$F = \begin{bmatrix} f_{1,1} & \cdots & f_{1,n} \\ \vdots & \ddots & \vdots \\ f_{128,1} & \cdots & f_{128,n} \end{bmatrix} \tag{21}$$

Step 2: Use Eq. (21)–(28) to calculate the parameters and initial values of the multi-time-vary NA-CML and LP map. These parameters and initial values are denoted as *Key*1.

$$x_{Lp} = \mod\left(\sum_{i=1,j=2}^{128} f_{i+1,j} f_{i,j}, 1\right) \tag{22}$$

$$x_{Lp} = \mod\left(\sum_{i=1,j=2}^{128} f_{i+1,j} f_{i,j}, 1\right) \tag{23}$$

$$x_{Lp} = \mod\left(\sum\nolimits_{i=1,j=2}^{128} f_{i+1,j} f_{i,j}, 1\right) \tag{24}$$

$$w = \begin{cases} floor(\mod(128 + (f_{1,i} \otimes f_{2,i} \ldots, \otimes f_{128,i}) \times 10^6, 100)), w < 2, i = [6, n] \\ floor(\mod(128 + (f_{1,5} \otimes f_{2,5} \ldots, \otimes f_{128,5}) \times 10^6, 100)), w \geq 2 \end{cases} \tag{25}$$

$$p, q = \begin{cases} floor\left(\mod\left(\dfrac{128 + (f_{1,j} \otimes f_{2,j} \ldots, \otimes f_{128,j}) \times 10^6}{128 \times 3}, 100\right)\right), p, q = 0, j = [i, n] \\ floor\left(\mod\left(\dfrac{128 + (f_{1,6} \otimes f_{2,6} \ldots, \otimes f_{128,6}) \times 10^6}{128 \times 3}, 100\right)\right), p, q \neq 0 \end{cases} \tag{26}$$

$$\varepsilon = \mod\left(\frac{\sum_{i=1}^{128} f_{i,j+1} f_{i+1,j+1}}{121}, 1\right) \tag{27}$$

$$\varepsilon = \mod\left(\frac{\sum_{i=1}^{128} f_{i,j+1} f_{i+1,j+1}}{121}, 1\right) \tag{28}$$

Step 3 Use DWT as a sparse basis Ψ to transform image into the frequency domain.Zigzag position transformation is performed on the transformed image to balance the overall sparsity. The changing way of Zigzag is shown in Fig. 6.

The Zigzag scrambled image is broken up into M $\sqrt{N} - \sqrt{N}$ non-overlapping sub-blocks. Each block is measured independently, with the definition of P_i being the i-th scrambled image block. To show the effect of encrypted images, we set the sampling rate to 0.5.

$$\begin{cases} \Psi_i = DWT(P_i) \\ P_i' = full(\Psi_i \times sparse(P_i) \times \Psi_i^T) \\ y_i = \Phi_i \times P_i' \end{cases} \tag{29}$$

y_i represents the measure values for the sub-block compressive sensing. The measuring matrix is Φ_i.

Step 4 Quantification of measure values using Eq. (10).

$$y_i = \lceil roundn(y_i, -5) \times 10000) \rceil \tag{30}$$

Step 5 The confusion sequence R is produced by iterating multi-time-vary NA-CML $n_0 + MN$ times using the parameters and initial values from Step 2.

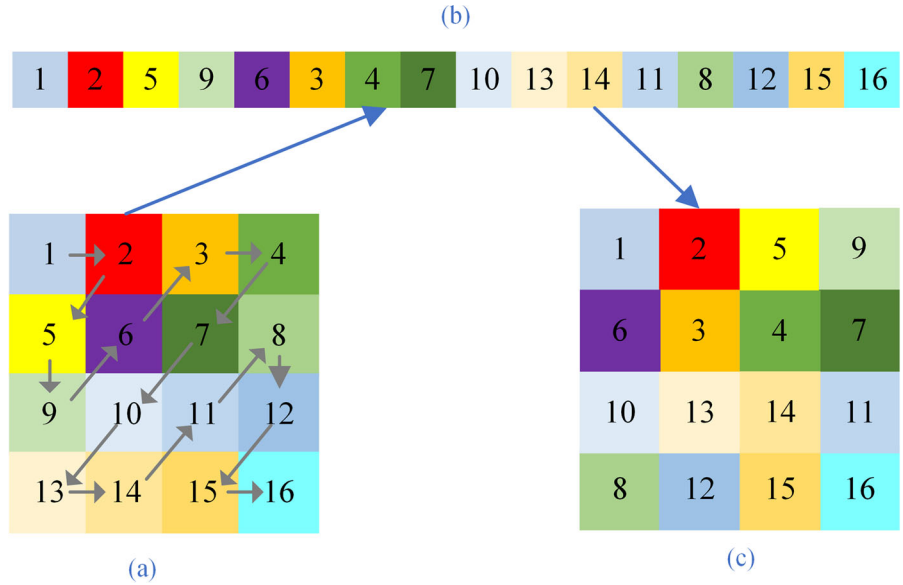


Fig. 6 ZigZag scramble process

Equation (32) is used to scramble the quantized values.

$$\begin{cases} R = NACML(Key1) \\ R = \begin{bmatrix} R_1 & \cdots & R_M \\ \vdots & \ddots & \vdots \\ R_N & \cdots & R_{MN} \end{bmatrix} \end{cases} \tag{31}$$

$$\begin{cases} cy = \mod(10^6 R + 6R + MN, NM) \\ c = joseScr(y, M/2, N, cy) \end{cases} \tag{32}$$

cy is the index of quantification values.

Step 6 The diffusion sequence is generated using Multi-time-vary NA-CML, with *Key*2 as the diffusion sequence's key. The NA-CML is iterated $n_0 + MN$ times, with the first n_0 values discarded after each iteration. The iterative values of NA-CML are the real number, and the diffusion operation cannot be performed directly. The iteration value needs to be converted to [0, 255]. Transform processes are shown in Eq. (4.13).

$$r(i) = \mod(round(r(i) \times 2^{14}), 256) \tag{33}$$

Use Eq. (34) to perform bit-XOR on the diffusion sequence to obtain the ciphertext image c.

$$\begin{cases} c_1 = y_1 \oplus r_1 \oplus y_{end}, i = 1 \\ c_i = y_i \oplus r_i \oplus c_{i-1}, i \neq 1 \end{cases} \tag{34}$$

The encryption result be shown in Fig. 7.

4.2 Decryption and reconstruction

After receiving the ciphertext images, the receiver sends the keys *Key*1 and *Key*2 to decrypt and reconstruct the ciphertext image through key negotiation with the sender. The decryption steps are as follows:

Step 1 Generate the quantized diffusion sequence using the *Key*1 according to Eq. (4.13). The inverse of the diffusion operation is performed on the ciphertext c using Eq. (4.15).

$$\begin{cases} y_1 = c_1 \oplus r_1 \oplus y_{end}, i = 1 \\ y_i = c_i \oplus r_i \oplus c_{i-1}, i \neq 1 \end{cases} \tag{35}$$

Step 2 Build Joseph's replacement subscript matrix cy with *Key*2, and carry out the Joseph confusion operation's inverse transformation on the ciphertext y.

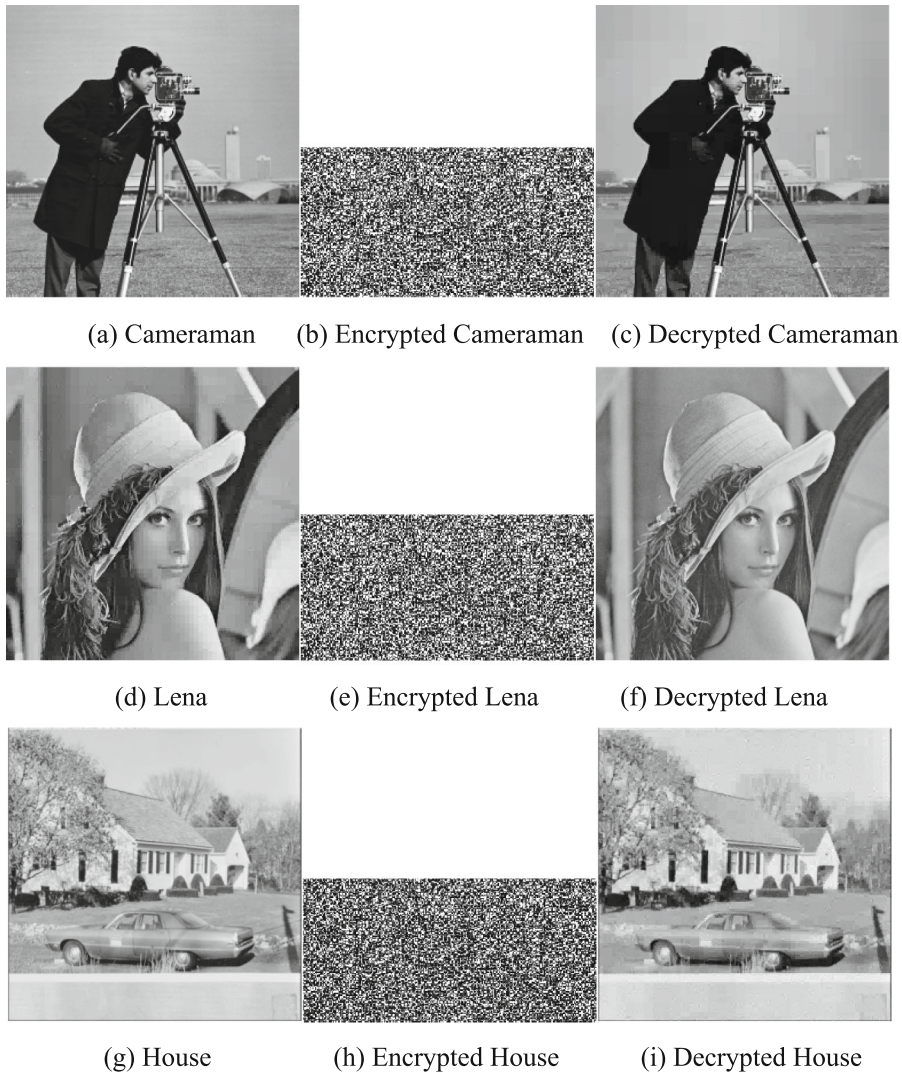
Step 3 The matrix y is inversely converted with quantized measurements, and the OMP is used to reconstruct the matrix y to get the reconstructed result P'.

Step 4 For P', the plaintext P is obtained using the inverse transform of the Zigzag method.

4.3 Encrypted image retrieval

This section describes the specific functions to retrieve encrypted images using the re-encryption technique,

Fig. 7 Encrypted results



(a) Cameraman (b) Encrypted Cameraman (c) Decrypted Cameraman

(d) Lena (e) Encrypted Lena (f) Decrypted Lena

(g) House (h) Encrypted House (i) Decrypted House

including key generate, key transform, index generate, trop door generate and search encrypted image. Meanwhile, the encrypted image is stored in Cloud1, while the index of the encrypted image is stored in Cloud2. The KMC is responsible for negotiating the keys needed for re-encryption. The retrieval functions are described as follows:

$(\varsigma, \Gamma, k, sk, pk) \leftarrow KeyGen(1^{\kappa})$: The key center input parameter κ, output bilinear pair parameter $\varsigma = (G, G_T, e, p, g)$ and secret parameter $\Gamma = \{s, A_1, A_2, \{h_k\}_{k=1}^{\lambda}, \{\psi_j\}_{j=1}^{L}, \{\varphi_j\}_{j=1}^{L}\}$, where s is a $d+1$ dimensional random invertible matrix, A_1 and A_2 are random invertible matrices of $(d+1) \times (d+1)$ dimension, $\{h_k\}_{k=1}^{\lambda}$ is λ hash functions, $\{\psi_j\}_{j=1}^{L}$ is a group of local sensitive hash functions, and $\{\varphi_j\}_{j=1}^{L}$ is the encryption function of L hash tables. The key generation center distributes the chaotic image encryption keys $k = \{k_i\}_{i=1}^{u}$ and the retrieval public and private key pair \$(sk, pk)\$ for the owner, satisfies $pk = g^{sk}$, assigns $\{sk_i, pk_i\}_{i=1}^{n}$ to the n users, and assigns the user with identity UID to the company key pair (sk_{UID}, pk_{UID}), satisfies $pk_{UID} = g^{sk_{UID}}$.

Algorithm 1 Key generation

Output: $(\varsigma, \Gamma, k, sk, pk)$

Input: κ

1. KMC generates keys encrypt key k and retrieval key pair (pk, sk)

2. **for** $\{sk_i, pk_i\}_{i=1}^{n}$ **do**

(sk_{UID}, pk_{UID}), **satisfies** $pk_{UID} = g^{sk_{UID}}$

3. **end for**

$\Lambda \leftarrow KeyTran(k, sk, pk)$: The image owner encrypts $k_i' = (pk^{\varepsilon_i}, k_i F^{\varepsilon_i})$ with the chaotic image encryption key k_i, where ε_i is a random number provided by the chaotic system under the current key, and F is a bilinear pair. Calculate the conversion key $TK_{UID} = g^{\frac{sk_{UID}}{sk}}$ for the user, then k_i corresponds to the re-encryption key $\Lambda_i = \{k_i', TK_{UID}\}$. Finally, the re-encryption key is sent to Λ to the Cloud2.

$I \leftarrow IndexGen(Feature, \Gamma)$: Locality Sensitive Hashing (LSH) is used to calculate the hash value of the *Feature* vector extracted by SIFT, and λ hash values $\psi(Feature_{i,t}) = \{h_k(Feature_{i,t})\}_{k=1}^{\lambda}$ are obtained. Next, $\psi()$ are applied to the eigenvectors to construct L hash tables. The function $\varphi()$ is used to encrypt the LSH bucket value $LSHKT_{ij}$, the secure nearest neighbor algorithm in [47] is used to encrypt the feature vector $\boldsymbol{f}$.

Algorithm 2 Index Generation

Output: I

Input: $Feature, \Gamma$

1. **for** $\psi(Feature_{i,t}) = \{h_k(Feature_{i,t})\}_{k=1}^{\lambda}$ **do**

2. $hash = LSH(Feature)$

3. $f = \varphi(LSHKT_{ij})$

4. **end for**

$TD \leftarrow TropDoor(\Gamma, Feature)$: The query user extracts the feature vector f_q of the query image, then use $\psi()$ to calculate *LSHKT*, and encrypts it with $\varphi(LSHKT)$. The query user extend f_q to $\overline{f}_q = (-2f_q, 1)$. For $l \in [1, d+1]$, if $s[l] = 0$, then the positive values of $\overline{f}_q^a$ and $\overline{f}_q^b$ are randomly choose, so that $\overline{f}_q^a[l] + \overline{f}_q^b[l] = \overline{f}_q[l]$ is satisfied. Otherwise $\overline{f}_q^a[l] = \overline{f}_q^b[l] = \overline{f}_q[l]$. Randomly select an positive number δ and multiply A_1^{-1}, A_2^{-1} by $\overline{f}_q^{aT}, \overline{f}_q^{bT}$, respectively, to obtain the encrypted query vector $f_q' = (\delta A_1^{-1}\overline{f}_q^{aT}, \delta A_1^{-1}\overline{f}_q^{bT})$. Finally, $\{\varphi(LSHKT_j)\}_{j=1}^{L}$, f_q' and UID are sent to the Cloud2 to form a trapdoor *TD*.

$R_q \leftarrow Search(TD, I, C, \Lambda)$: After receiving the *TD*, the Cloud2 searches the index table for the bucket with the same value as the trapdoor bucket, and the images in the bucket can be used as candidate images. Then, Eq. (36) is used to calculate the inner product value of the feature vector of the candidate image set and the query feature vector.

$$\begin{aligned} f_q^{T'} f_{i,t}' &= (\delta A_1^{-1}\overline{f}_q^{aT})^T A_1^T \overline{f}_{i,t}^{aT} + (\delta A_2^{-1}\overline{f}_q^{bT})^T A_2^T \overline{f}_{i,t}^{bT} \\ &= \delta \overline{f}_q \overline{f}_{i,t}^T = \delta(||f_{i,t}||^2 - 2f_q f_{i,t}^T) \\ &= \delta(||f_q - f_{i,t}||^2 - ||f_q||^2) \end{aligned} \tag{36}$$

The nearest image is selected according to the size of the inner product value and returned to the query user. The Cloud2 calculates $e(pk^{\varepsilon_i}, TK_{UID}) = e(g^{sk_{UID}}, g^{\frac{sk_{UID}}{sk}}) = e(g,g)^{\varepsilon_i sk_{UID}} = F^{\varepsilon_i sk_{UID}}$ based on the ciphertext of the key k_i' corresponding to the encrypted image and the transformation key TK_{UID} associated with the user UID. Then Cloud1 and Cloud2 return the corresponding re-encryption key as the calculation result to the query user, and the users decrypt it with query user legal private key to obtain the decryption key.

5 Experimental analysis

Our solution's encryption, decryption, and encrypted image retrieval functions are implemented on Windows OS with Intel(R) Core(TM) i5-8265U CPU @ 1.60GHz using MATLAB 2022b and Java.

5.1 Key analysis

The key space must be sufficiently large to guarantee that the encryption system can withstand brute-force attacks [48]. *Key*1 and *Key*2 are computed using the eigenvalues of the plaintext images. The constants in each parameter are made up of real integers, and the keys accept values between $[0, +\infty]$. Therefore, the space of our proposed key system is large enough to resist the analysis of brute force attacks.

Key sensitivity indicates that even a slight change in the key cannot decrypt the ciphertext images. We added 10^{-10} to the feature vector to generate new *Key*1 and *Key*2 to decrypt the ciphertext image. The decrypted results are displayed in Fig. 8b–d. We randomly choose a pixel in the original image to modify only 1 bit, using the same key to encrypt two images; the difference between the two encrypted images is shown in Fig. 8e.

5.2 Histogram and its related analysis

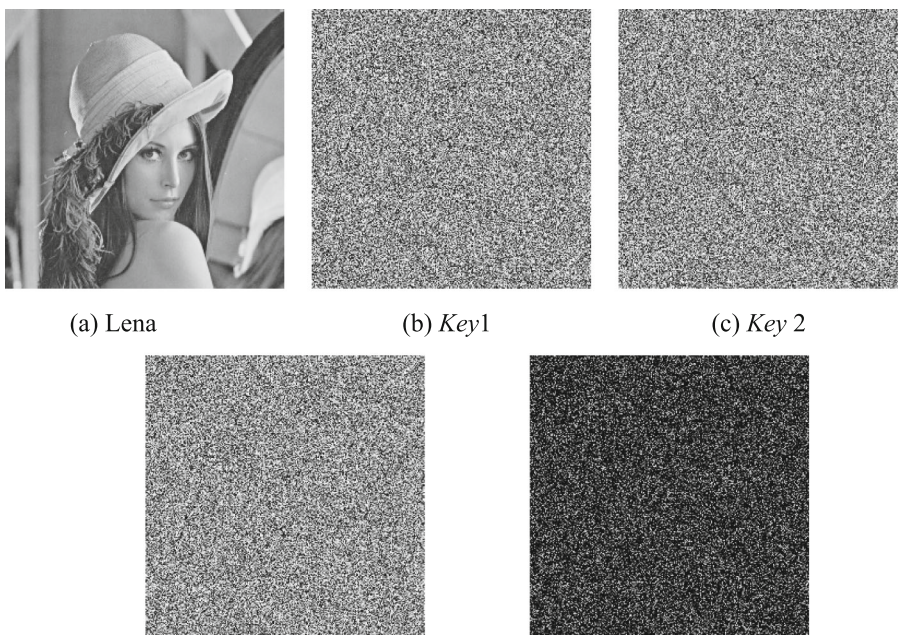
The histogram visually describes the pixel distribution of images. The histogram of a plaintext image is uneven, representing this image feature. The attacker can analyze the distribution of the histogram and then obtain plaintext images. Therefore, the histogram of ciphertext images should be uniformly distributed and cannot be analyzed by an attacker for distribution. We use χ^2 test to quantify the distribution of the encrypted image. Figure 9 displays the histograms of the cameraman, Lena, and House before and after encryption, respectively. Table 1 displays the χ^2 test results.

The mathematical definition of the χ^2 test is as follows:

$$\chi^2 = \sum_{i=0}^{255} \frac{(p_i - \overline{p})^2}{\overline{p}} \tag{37}$$

The p_i is the value of the ith pixel, $\overline{p} = (M \times N)/256$. The criterion for judging the uniformity of the histogram is $\chi^2_{0.05} \leq 293.24783$ when the significant level α is set to 0.05.

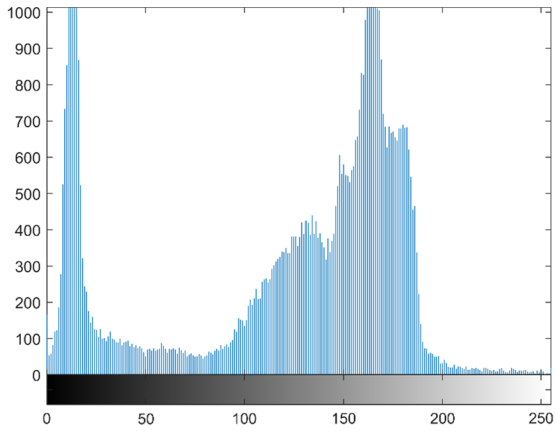
The encrypted image's pixel distribution appears relatively uniform in Fig. 9, and the quantized values in Table 1 pass the χ^2 test, proving the resistance of our proposed algorithm to the statistics attack analysis.



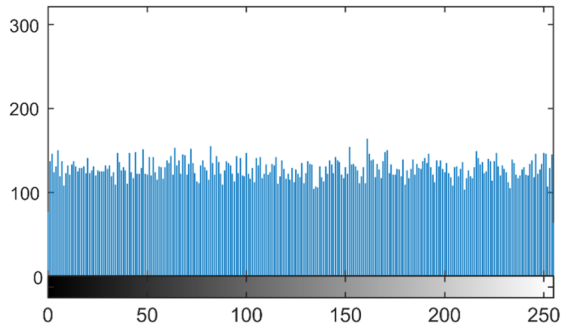
(a) Lena (b) *Key*1 (c) *Key* 2

(d) 1-bit (e) Diff

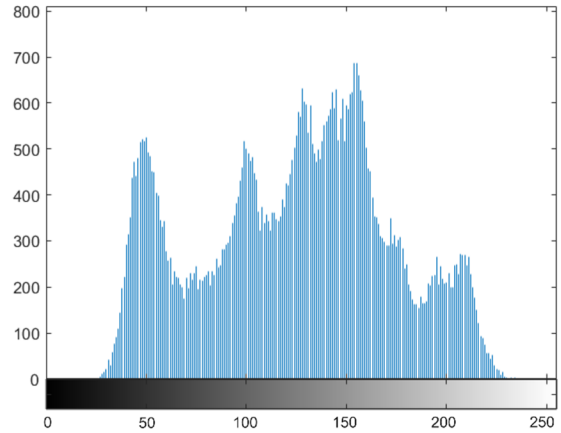
Fig. 8 Key sensitivity test



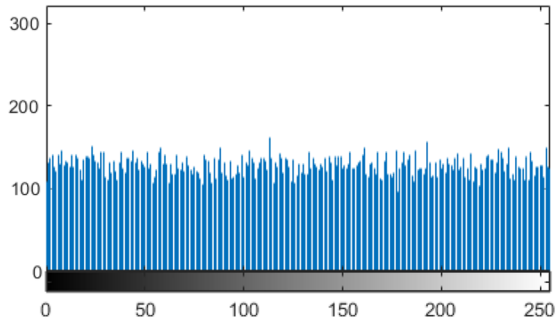
(a) histogram of cameraman



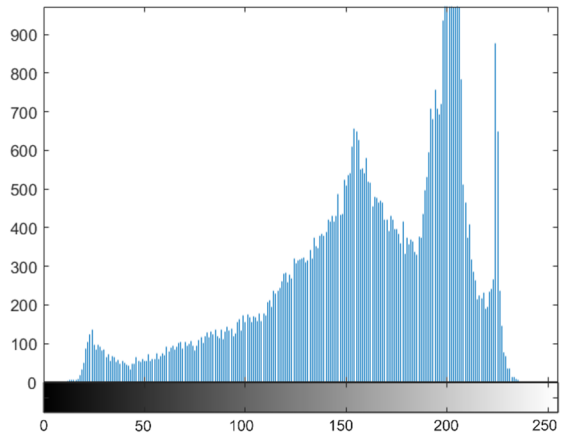
(b) histogram of ciphertext cameraman



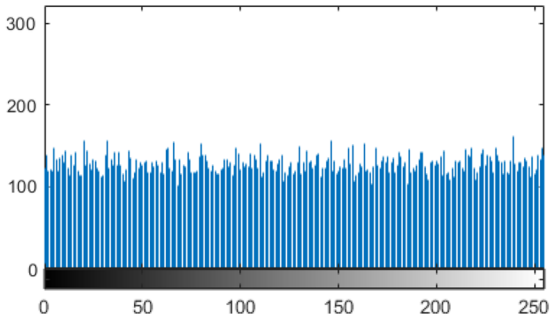
(c) histogram of Lena



(d) histogram of ciphertext Lena



(e) histogram of house



(f) histogram of ciphertext house

Fig. 9 Histograms

Table 1 χ^2 test

	Plaintext	Encrypted Image	Pass/Fail
Lena	27,874	237.7656	Pass
Cameraman	98,781	266.9531	Pass
House	78,604	199.1719	Pass

5.3 Information entropy analysis

The information entropy value is closer to 8, theoretically representing better resistance to statistical attacks. The mathematical definition of information entropy is shown Eq. (38):

$$H(s) = \sum_{i=0}^{2^{N-1}} p(s_i) \log_2 \frac{1}{p(s_i)} \tag{38}$$

where $p(s_i)$ represents the probability of occurrence of s_i.

It can be seen from Table 2 that the information entropy of the encrypted image is very close to the ideal value of 8. It proves that the encrypted image has been completely confused.

5.4 Local information entropy

The local information entropy measures the degree of confusion of the local ciphertext images. The mathematical definition of local information entropy is shown in Eq. (39):

$$H^{-}_{(k,T_B)(S)} = \sum_{i=1}^{k} \frac{H(S_i)}{k} \tag{39}$$

S_i is the size of the randomly chosen blocks ciphertext image, T_B is the number of pixels in the S_i block, and k is the index of randomly chosen blocks. The information entropy of this block is $H(S_i)$. According to [23], the local information entropy should be in the range [7.901722822,7.903215812] with $T_B = 1936$, $k = 30$ and confidence level $\alpha = 0.001$. The Local information entropy of the encrypted images in Table 3 can all pass the test.

From Table 3, we can see that our proposed scheme can pass the local information entropy test, proving that our proposed scheme's local confusion level is also excellent.

Table 2 Information entropy comparison

Image	Plain	Proposed
Lena	7.44409	7.99964
Cameraman	7.10471	7.99933
House	7.22879	7.99986
5.1.09	6.709312336	7.99808
5.1.10	7.311807223	7.99886
5.1.11	6.452275232	7.998443
5.1.12	6.705667201	7.999353
5.1.13	1.548313697	7.99962
5.1.14	7.342432513	7.99642
5.2.08	7.20100796	7.99963
5.2.09	6.993994166	7.99834
5.2.10	5.705560158	7.99935
7.1.01	6.027414821	7.998731
7.1.02	4.004499445	7.997982
7.1.03	5.495739989	7.997746
7.1.04	6.107418111	7.997727
7.1.05	6.563195556	7.999770
7.1.06	6.695283381	7.997332
7.1.07	5.991598763	7.998209
7.1.08	5.053447997	7.999129
5.3.01	7.523736852	7.999312
5.3.02	6.83032952	7.999754
7.2.01	5.641453727	7.999301

5.5 Differential attack analysis

The differential attack is to encrypt two plaintexts with only minor differences using the same key and observe the change of the ciphertext images. If the pixel change of the two ciphertext images is tremendous, it proves that the avalanche effect of the encrypted algorithm is excellent. In this paper, we use the number of pixel change rates (NPCR) and unified average changing intensity (UACI) to measure the resistance of differential attacks. Theoretically, the closer the NPCR is to 99.6093% the closer the UACI is to 33.4635%, the better. he mathematical definition of NPCR and UACI are shown in Eqs. (40)–(42):

$$NPCR = \frac{\sum_{i,j} D(i,j)}{W \times H} \times 100 \tag{40}$$

Table 3 Local Information Entropy

Image	Proposed	Pass/Fail
Lena	7.90254776226	Pass
Cameraman	7.90293592075	Pass
House	7.90304617642	Pass
5.1.09	7.901485587	Pass
5.1.10	7.903201336	Pass
5.1.11	7.903292228	Pass
5.1.12	7.901771395	Pass
5.1.13	7.903166549	Pass
5.1.14	7.90188768	Pass
5.2.08	7.903210092	Pass
5.2.09	7.903051248	Pass
5.2.10	7.902697475	Pass
7.1.01	7.901380099	Pass
7.1.02	7.901195181	Pass
7.1.03	7.90288683	Pass
7.1.04	7.903059891	Pass
7.1.05	7.903036833	Pass
7.1.06	7.902121419	Pass
7.1.07	7.902376528	Pass
7.1.08	7.901820954	Pass
5.3.01	7.903022515	Pass
5.3.02	7.902883762	Pass
7.3.01	7.903142077	Pass

$$UACI = \frac{1}{W \times H}\left[\sum_{i,j}\frac{|c_1(i,j) - c_2(i,j)|}{255}\right] \times 100 \tag{41}$$

$$\begin{cases} D(i,j) = 1, if\ c_1(i,j) \neq c_2(i,j) \\ D(i,j) = 1, if\ c_1(i,j) = c_2(i,j) \end{cases} \tag{42}$$

W and H are the width and height of the image, respectively. c_1 and c_2 are the two ciphertext images, respectively.

The [49] proposed a statistical hypothesis model for NPCR and UACI under ideal conditions, and the mathematical definitions of non-significant NPCR and UACI are as follows:

The result of non-significant NPCR and UACI are shown in Tables 4 and 5.

$$\begin{cases} H_0 : NPCR(c_1, c_2) = \mu_N \\ H_1 : NPCR(c_1, c_2) < \mu_N \end{cases} \tag{43}$$

$$NPCR^*_\alpha = \frac{F - \Phi^{-1}(\alpha)\sqrt{\frac{F}{MN}}}{F+1} \tag{44}$$

$$\begin{cases} H_0 : UACI(c_1, c_2) = \mu_U \\ H_1 : UACI(c_1, c_2) < \mu_U \end{cases} \tag{45}$$

$$\begin{cases} UACI^{*-}_\alpha = \frac{F+2}{3F+3} - \Phi^{-1}(\alpha)\sqrt{\frac{(F+2)(F^2+2F+3)}{18(F+1)^2MNF}} \\ UACI^{*+}_\alpha = \frac{F+2}{3F+3} + \Phi^{-1}(\alpha)\sqrt{\frac{(F+2)(F^2+2F+3)}{18(F+1)^2MNF}} \end{cases} \tag{46}$$

5.6 Correlation analysis

The capacity to break the association between adjacent image pixels is essential for the encryption algorithm. In this paper, 10,000 pairs of pixels are randomly selected in the horizontal, vertical, and diagonal directions of the plaintext and ciphertext images of Lena to test the correlation of the pixels in these three directions. The test results are shown in Fig. 10.

In addition, we use the correlation coefficient to measure the correlation between the adjacent pixels. The calculation method of the correlation coefficient is shown in Eq. (5.11). The results we show in Table 6. The experimental findings clearly demonstrate the presence of a strong correlation among neighboring pixels in the original plaintext image. However, upon employing the algorithm proposed in this paper for encryption, the correlation between adjacent pixels in the ciphertext image significantly diminishes, resulting in a much smaller correlation compared to that observed in the Lena plaintext image. Consequently, it can be concluded that the suggested algorithm effectively withstands statistical analysis attacks.

Table 4 NPCR

Size	Image	NPCR			
		Our	[15]	[50]	[43]
256 × 256	Camera man	99.5727	–	–	–
$NPCR^*_\alpha \geq 99.5693$	Lena	99.6282	–	–	–
	House	99.6719	–	–	–
	5.1.09	99.6544	99.6277	99.6170	99.5895
	5.1.10	99.5962	99.5956	99.6170	99.6353
	5.1.11	99.6075	99.5758	99.5529	99.6368
	5.1.12	99.6625	99.6323	99.6033	99.5987
	5.1.13	99.6823	99.6033	99.6033	99.6109
	5.1.14	99.5799	99.6277	99.5636	99.5819
512 × 512	5.2.08	99.6051	99.5968	99.6067	99.5934
$NPCR^*_\alpha \geq 99.5893$	5.2.09	99.6855	99.6052	99.6151	99.6002
	5.2.10	99.6354	99.6039	99.5987	99.6365
	7.1.01	99.6275	99.6090	99.5918	99.6147
	7.1.02	99.5901	99.5961	99.6124	99.6075
	7.1.03	99.6541	99.6071	99.6346	99.6048
	7.1.04	99.6358	99.6326	99.6029	99.6326
	7.1.05	99.6056	99.6170	99.6063	99.6086
	7.1.06	99.6215	99.6193	99.6262	99.6185
	7.1.07	99.6438	99.6056	99.5953	99.6008
	7.1.08	99.5994	99.6124	99.6128	99.5958
1024 × 1024	5.3.01	99.6046	99.6033	99.6115	99.6104
$NPCR^*_\alpha \geq 99.5994$	5.3.02	99.6588	99.6094	99.6077	99.6096
	7.3.01	99.6417	99.6178	99.6142	99.6066
	Mean	**99.62889**	**99.6099**	**99.60467**	**99.60966**
	Pass/All	23/23	23/23	23/23	23/23
	Std	0.031559	0.013712	0.018323	0.015341

$$\begin{cases} r_{xy} = \dfrac{\operatorname{cov}(x,y)}{\sqrt{D(x)}\sqrt{D(y)}} \\ \operatorname{cov}(x,y) = \dfrac{1}{N}\sum_{i=1}^{N}(x_i - E(x))(y_i - E(y)) \\ D(x) = \dfrac{1}{N}\sum_{i+1}^{N}(x_i - E(x))^2 \\ E(x) = \dfrac{1}{N}\sum_{i=1}^{N} x_i \end{cases} \tag{47}$$

5.7 Robustness analysis

5.7.1 Noise attack analysis

Images are easily disturbed by the environment during transmission, making it impossible to decrypt ciphertext images correctly. In this paper, we choose different intensities of salt-pepper noise to decrypt the contaminated encrypted images. We add the pepper noise with the intensity of 0.0001,0.001, and 0.01, respectively, to the cipher image, and the decryption results are in Fig. 11. From Fig. 11, we can see that the decryption results can be distinguished by the naked eye even if they are affected by the salt-pepper noise.

5.7.2 Crop attack analysis

The ciphertext is easily disturbed by the environment during transmission, making the encrypted image lose data. In Fig. 12, the decryption effect of losing 8 × 8, 16 × 16, 32 × 32 data is shown. Figure 12 shows that the decryption effect after the cropping attack deteriorates the image quality. However, it can still be

Table 5 UACI

Size	Image	UACI			
		Our	[15]	[50]	[43]
256×256	Camera man	33.4905	–	–	–
$(UACI_{\alpha}^{*-}, UACI_{\alpha}^{*+}) = (33.2824, 33.6447)$	Lena	33.5698	–	–	–
	House	33.4127	–	–	–
	5.1.09	33.5564	33.5953	**33.6739**	33.4605
	5.1.10	33.4531	33.4032	33.5039	33.3595
	5.1.11	33.4505	33.5592	33.5073	33.5051
	5.1.12	33.6054	33.4884	33.4536	33.3532
	5.1.13	33.5865	33.5461	33.3681	33.3641
	5.1.14	33.5940	33.3877	33.6094	33.4708
512×512	5.2.08	33.4238	33.4637	33.5379	33.4642
$(UACI_{\alpha}^{*-}, UACI_{\alpha}^{*+}) = (33.3730, 33.5541)$	5.2.09	33.3708	33.4255	**33.3704**	33.5235
	5.2.10	33.5410	33.4695	33.5167	33.4638
	7.1.01	33.5238	33.4884	33.5084	33.4310
	7.1.02	33.4299	33.4544	33.4485	33.5172
	7.1.03	33.4016	33.4751	33.5236	33.4884
	7.1.04	33.5416	33.4960	33.5454	33.4156
	7.1.05	33.5304	33.4984	33.3958	33.4123
	7.1.06	33.5123	33.3944	33.4756	33.5102
	7.1.07	33.5241	33.4850	33.4469	33.4759
	7.1.08	33.5017	33.3823	33.5569	33.4500
1024×1024	5.3.01	33.4664	33.4632	33.5039	33.4341
$(UACI_{\alpha}^{*-}, UACI_{\alpha}^{*+}) = (33.4183, 33.5088)$	5.3.02	33.4921	33.4521	33.4856	33.4485
	7.3.01	33.5013	33.4743	33.4813	33.4418
	Mean	**33.49912**	**33.47011**	**33.495655**	**33.449485**
	Pass/All	23/23	23/23	21/23	23/23
	Std	0.126319	0.054382561	0.07189782	0.048928737

decrypted, proving that our algorithm can resist the cropping attack.

5.8 Encrypted image retrieval accuracy

To test the retrieval accuracy of encrypted images, we use Caltech 101 dataset as the retrieval experiment object, which contains 102 classes. We randomly select ten categories in Caltech 101 dataset as the retrieval data sets and repeat three times to calculate the retrieval accuracy of encrypted images under top-k.

From Fig. 13a–c, we can conclude that the retrieval accuracy of our proposed algorithm is higher than the accuracy of CSD, CLD, and EHD of the [28, 51].

5.9 Efficiency analysis

In addition, to illustrate the computational efficiency of our algorithm, we give the time of encrypting the image, index construction and retrieving the image for comparison with other similar algorithms in Tables 7 and 8.

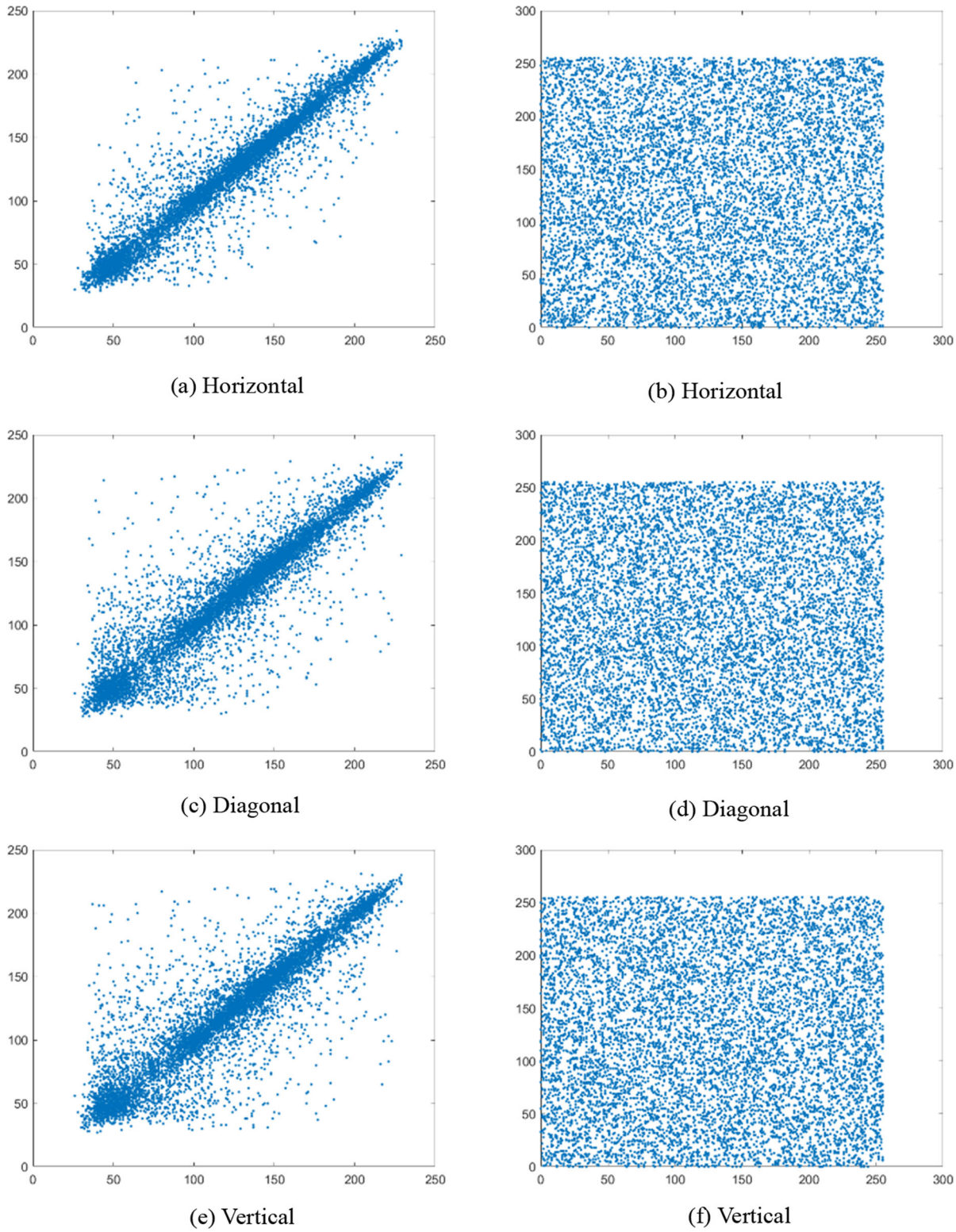


Fig. 10 Correlation

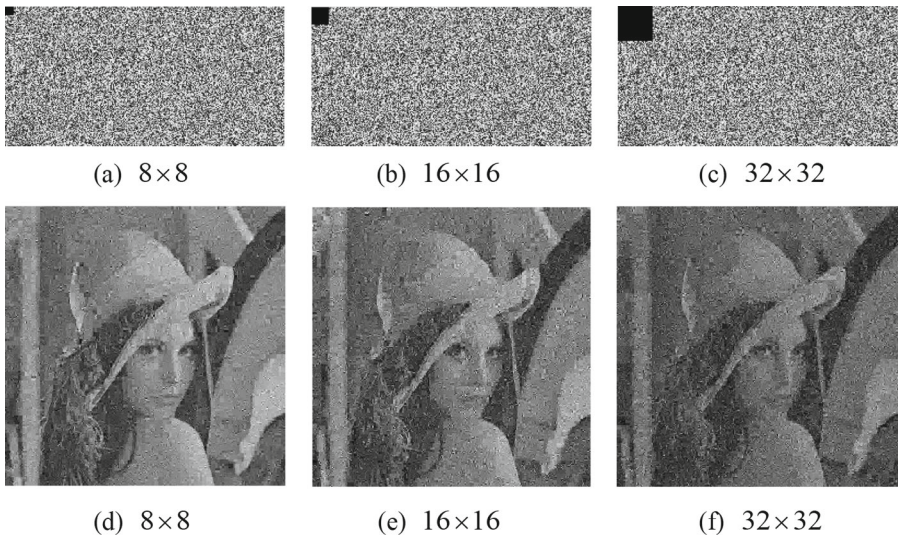
Table 6 Correlaation between adjacent pixels

	Plaintext image			Ciphertext Lena		
	Horizontal	Vertical	Diagonal	Horizontal	Vertical	Diagonal
Cameraman	0.9676	0.9685	0.9714	0.0450	0.0728	-0.1127
Lena	0.9916	0.8237	0.9518	0.0010	-0.0452	0.0056
House	0.9926	0.8916	0.9330	-0.0023	0.0128	-0.1327



(a) 0.0001 (b) 0.001 (c) 0.01

Fig. 11 Decrypted images with different intensity nois



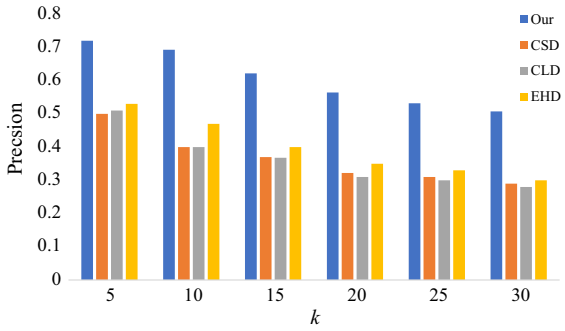
(a) 8×8 (b) 16×16 (c) 32×32

(d) 8×8 (e) 16×16 (f) 32×32

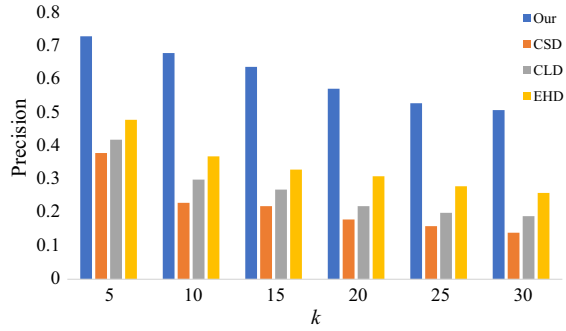
Fig. 12 Decrypted image with different data lost

From Table 7, we can see that the average encryption and decryption times for 256 × 256, 512 × 512, and 1024 × 1024 images are 2.62s, 4.93s, and 11.09s. The average decryption times are 2.84s, 3.89s, and 19.88s. Our proposed scheme is not as good as the [15, 52, 53] in terms of encryption and decryption time, but the encryption and decryption times are still within the acceptable range.

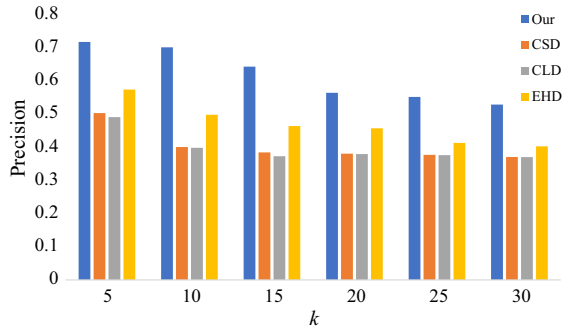
Also, to illustrate the superiority in building indexes, we tested the index building time on Test1,



(a) Test1



(b) Test2



(c) Test3

Fig. 13 Average retrieval precision for different test sets when retrieving top-*k* results

Table 7 Encryption/decryption time comparison

	Our method	[15]	[52]	[53]
256*256	2.62s	0.12s	0.66s	0.44s
	2.84s	0.12s	–	–
512*512	4.93s	0.51s	1.06s	2.77s
	3.89s	0.50	–	–
1024*1024	11.09s	–	–	–
	19.88s	–	–	–

Table 8 Comparison of image index construction time

Images	Test 1	Test 2	Test 3	[2]	[54]
100	2	3	1.2	48	72
200	3.5	5.2	4	62	146
300	4.3	6.9	4.3	104	227
400	6.8	8.8	6.8	156	303
500	8.4	11	7.5	182	379
800	12	15	12	201	437

Test2, and Test3, compared it with similar study [2, 54] and found that our building index time is less, compared result in Table 8.

6 Conclusion

To solve the privacy leakage problem and encrypted images cannot be retrieved, we propose new full chaos Multi-time-vary NA-CML and its application in encrypted image retrieval, ensuring that privacy is not leaked in both upload and retrieval phases. In the encryption phase, we introduce time-varying functions and UKF to improve NA-CML system complexity and security and use the image feature vector as the encryption key can achieve the ideal encryption target of the one-time pad. In the retrieval phase, we use the sensitive hash to reduce the feature vector's dimensionality and improve the efficiency of constructing the index. We perform a secondary hash on the reduced data to form a new hash-key retrieval structure, improving retrieval efficiency. When designing the system architecture, we assume that the cloud server is semi-honest, so the retrieval vector and the encrypted image are stored separately to protect user privacy from leakage. KMC is also introduced to implement the retrieval of encrypted image retrieval. The experimental results show that our proposed fully chaotic system is better in chaotic degradation and security than the one-dimensional chaotic system, less vulnerable to cracking, and easier to be implemented in engineering. Our proposed retrieval scheme has a nominal communication cost in multi-key (user) scenarios and is closer to real-world needs.

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Data availability Data available on request from the authors. The data that support the findings of this study are available from the corresponding author, upon reasonable request.

Declaration

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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